Bae Jinwoo

AI Researcher & Engineer | bgw4399@gmail.com | github.com/bgw4399 | Kaggle: bgw4399

AI engineer focused on turning unfamiliar domain data into working systems. Experienced in medical AI, clinical data pipelines, machine unlearning research, Kaggle competitions, CAD/DXF-based engineering tools, and data-driven operation dashboards. Strengths include fast domain learning, domain-to-system translation, reproducible experimentation, and cross-functional communication.

EDUCATION

Kookmin University - B.S. in Artificial Intelligence

Mar 2022 - Feb 2026

- Relevant coursework: Data Structures, Applied Statistics, Computer Vision, Deep Learning, Database, Software Architecture, AI Hardware.

RESEARCH EXPERIENCE

Korea University College of Medicine - Clinical AI Research

Oct 2025 - Present | Collaborating Researcher

- Researching electrolyte response and fluid treatment prediction for MICU hyponatremia patients.
- Contributing to variable definition, missing-data interpretation, feature-set design, model evaluation, and explainability review.

Kim Kwangsoo Lab - Medical AI and Clinical Data Systems

Nov 2024 - Dec 2025 | AI Researcher

- Worked on clinical data, OMOP CDM workflows, pathology image registration, LightGlue/SuperGlue matching, and cell segmentation.
- Built Bento/CDM Explorer AI components that convert natural language or paper-derived clinical criteria into structured cohort conditions.
- Implemented an MCP-style literature and patient-feature extraction flow for protein-variant-related research support.

Kookmin University ML&PR Lab - Machine Unlearning Research

Aug 2023 - Feb 2024 | Undergraduate Researcher

- Researched gradient-similarity-based weight initialization for machine unlearning; accepted to PoPETs 2026.
- Ran pruning, knowledge distillation, poisoning, CIFAR/SVHN, retain/forget utility, and privacy-metric experiments.

SELECTED PROJECTS

Bento/CDM Explorer Medical CDM research platform; AI pipeline for cohort-condition extraction from natural language and papers.	NVIDIA Nemotron Challenge LoRA/PEFT reasoning adapters, verifier audits, model diversity, TIES/DARE/model-soup strategy.
NeuroGolf 2026 Task-level ONNX profiling, compact graph-surgeon blending, score-cost audit; tracked score up to 6405.61.	Electrical DWG/DXF App FastAPI CAD estimation and routing tool with DXF coordinate unification, overlays, Hanan Grid/Steiner routing.
Cosmetics Sample System Excel-to-SQLite operations dashboard with generated reports and rule-based question answering.	Stock News Analysis Desk Local news/price dashboard that classifies news impact and preserves source evidence for review.
CAFA6 Protein Function Prediction ESM/ProtBERT-style embeddings, ontology features, stacking, XGBoost/scikit-learn pipelines.	Pathology Image AI H&E/IHC registration, feature matching, preprocessing and annotation-quality review for segmentation workflows.

AWARDS AND PUBLICATIONS

- PoPETs 2026 Accepted Paper - Machine unlearning with gradient-similarity-based weight initialization.
- NeurIPS 2023 Machine Unlearning Challenge, 2nd Place - Kaggle / NeurIPS 2023.
- Best Poster Presentation Award - Korea University International Medical Student Research Conference, 2025.
- Best Poster Award - Korean Society of Medical Informatics Fall Conference, 2025.
- Capstone Design Excellence Award - Kookmin University Software Convergence College, 2025.

SKILLS

AI/ML	PyTorch, scikit-learn, XGBoost, LoRA/PEFT, LLM fine-tuning, model merging, computer vision, medical AI.
Data/System	Python, FastAPI, SQLite, pandas, NumPy, PowerShell, JavaScript, HTML/CSS, REST APIs, report generation.
Domain Tools	OMOP CDM, pathology image registration, CAD/DXF processing, route optimization, MCP/agent workflows.
Working Style	Domain translation, reproducible experiments, cross-functional communication, evidence-based iteration.